# Week 14: From Research to Impact: Intellectual Property, Technology Transfer and Spin-offs

## R&D and Project Management in Computer Science

R&D and Project Management in Computer Science (11117BLG002). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

## Overview

Research creates value when its results are used. This week examines models of knowledge transfer and the protection of intellectual property in computing. It then turns to the institutions of technology transfer in Türkiye and to the decision between licensing a technology and founding a spin-off company [1, 2, 3].

## Learning outcomes

By the end of the week, students are expected to describe the triple helix and open innovation models. They should be able to explain the requirements for patents and the limits on patenting software, and to identify the institutions of technology transfer in Türkiye. Finally, they should be able to compare licensing and spin-off options with net present value under uncertainty.

## Lecture

### Models of knowledge transfer

Etzkowitz and Leydesdorff described the relations among universities, industry and government as a triple helix [1]. In a triple helix, each sphere takes on some roles of the others, for example when universities found companies and governments act as venture investors. Chesbrough argued that firms increasingly use open innovation, combining internal and external ideas and paths to market, which creates demand for university research and for licensing [2]. Rogers's study of the diffusion of innovations showed that adoption depends on characteristics such as relative advantage, compatibility, complexity, trialability and observability [4]. These characteristics are a useful checklist for the impact section of a proposal.

## Protecting results

A patent protects an invention that is new, involves an inventive step and is susceptible of industrial application. Under the European Patent Convention, programs for computers as such are excluded from patentability, although inventions that solve a technical problem with software can be patented [5]. Türkiye's Industrial Property Law No. 6769 of 2016 follows the same approach and contains specific provisions on inventions made at higher education institutions [6]. Novelty is lost when an invention is published before a patent application is filed. Researchers who intend to patent must therefore coordinate with their technology transfer office before submitting papers or presenting results. Software code itself is protected by copyright and can be licensed, including under open-source licences. In the United States, the Bayh-Dole Act of 1980 allowed universities to own inventions from federally funded research. Mowery and colleagues found that university patenting and licensing grew in this period, but that the Act was only one of several causes [7].

## Technology transfer in Türkiye

Most universities in Türkiye have technology transfer offices that advise on intellectual property, contracts with industry and funding. Technology development zones under Law No. 4691 host research-based companies, including academic spin-offs, in or near universities [8]. TÜBİTAK's 1512 programme supports entrepreneurs in turning technology ideas into companies, and the 1505 programme supports university-industry projects. The route from a research result to a product usually passes through several of these institutions. Early contact with the technology transfer office avoids mistakes such as premature publication or unclear ownership.

## Licensing or a spin-off?

Shane studied university spin-offs and showed that they arise mainly from inventions that are radical, tacit and at an early stage, for which licensing to established firms is difficult [3]. Ries's lean start-up method recommends testing the key assumptions of a business with a minimum viable product in cycles of building, measuring and learning, before large investments [9]. The financial comparison of licensing and a spin-off can be framed with net present value. Licensing yields smaller but more certain income, while a spin-off requires investment and yields large returns only if it succeeds. Figure 14.1 shows the result of a simulation under illustrative assumptions, and the interactive lab lets students vary them. Such calculations do not replace judgement, but they make assumptions explicit and show how sensitive the decision is to the probability of success.

> ***Pause and reflect.*** *Which result of your project could be used outside academia, and who would have to adopt it? What would have to be true for a spin-off to be the better route?*

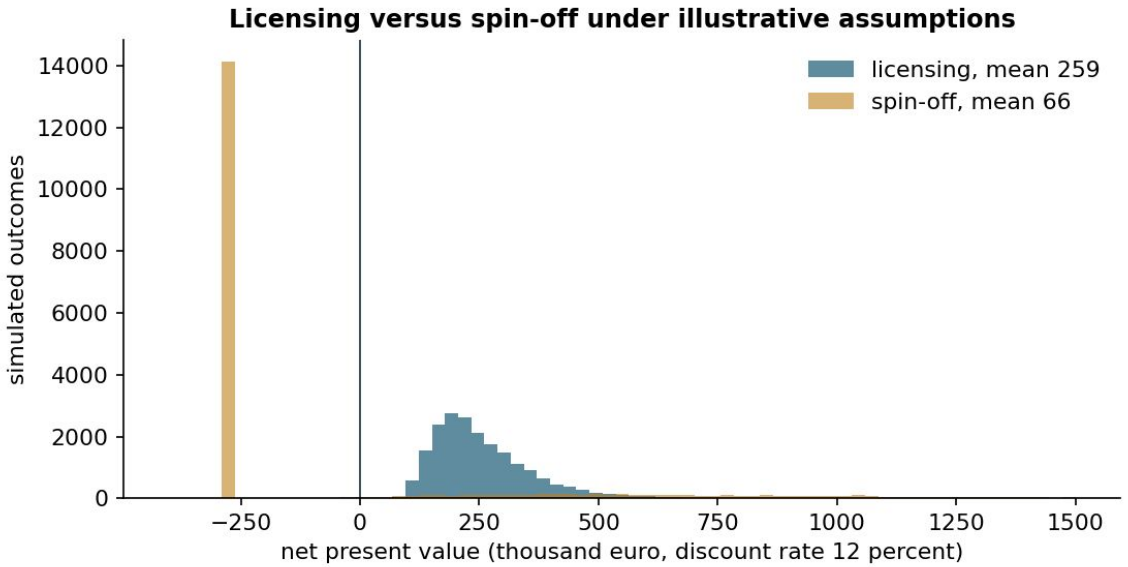



*Figure 14.1. Simulated net present values of licensing and of founding a spin-off for the same technology, under illustrative assumptions. Licensing is safer, while the spin-off has a small chance of a much larger return.*

## Hands-on components

Interactive lab: Part A suggests forms of protection for a research result from a few questions; it is an orientation aid, not legal advice. Part B compares licensing with a spin-off by simulation of net present value under assumptions that you set [3, 5]. Link: https://utkukose.github.io/rd-project-management-NB-lecture/weeks/week-14/lab.html

Colab notebook: It implements net present value, simulates licensing and spin-off outcomes under illustrative assumptions and compares their expected values and risks [3]. Link: https://colab.research.google.com/github/utkukose/rd-project-management-NB-lecture/blob/main/weeks/week-14/NB14_research_to_impact.ipynb

## Reflection prompts

1. Which result of your project could be protected, and in what form? Who would you contact first at your institution?
2. At what probability of success did the spin-off become better in the interactive lab? How confident are you in that probability?
3. Rewrite the impact pathway of your proposal with Rogers's adoption characteristics in mind.

## Weekly task and submission

**What to do.** Write the exploitation and impact plan of your project, about 700 words. First, list the results that others could use. Second, state how each result would be protected or shared, for example by a patent, copyright or an open licence. Third, choose a route to users, such as

licensing, a spin-off or an open release, and justify it. Fourth, compare two routes with a simple NPV calculation whose assumptions are stated explicitly. Finally, combine your midterm proposal, revised if needed, with the plans of Weeks 8 to 14 into one proposal portfolio.

**Where to look.** Lecture: sections "Protecting results", "Technology transfer in Türkiye" and "Licensing or a spin-off?". Interactive lab: Part A, which suggests forms of protection, and Part B, which compares licensing with a spin-off. Colab notebook: Section 1 with Exercise 1 for NPV and Section 2 with Exercise 2 for the simulated comparison.

**Example.** A trained model and its software could be released under an open licence, while a new hardware sensor could be patented. If the software is the main result, an open release with paid support may serve users better than a licence. The NPV comparison states the discount rate, the time horizon and the probability of success.

**What to submit.** The plan of about 700 words with the NPV table, the proposal portfolio as one document, and the Colab notebook with Exercises 1 and 2 completed. This weekly task is separate from the final capstone, whose page lists its own deliverables.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

## References

[1] Etzkowitz, H., & Leydesdorff, L. (2000). The dynamics of innovation: From National Systems and 'Mode 2' to a Triple Helix of university-industry-government relations. Research Policy, 29(2), 109-123. https://doi.org/10.1016/S0048-7333(99)00055-4

[2] Chesbrough, H. W. (2003). Open Innovation: The New Imperative for Creating and Profiting from Technology. Harvard Business School Press.

[3] Shane, S. (2004). Academic Entrepreneurship: University Spinoffs and Wealth Creation. Edward Elgar.

[4] Rogers, E. M. (2003). Diffusion of Innovations (5th ed.). Free Press.

[5] European Patent Office (2020). European Patent Convention, Article 52: Patentable inventions. https://www.epo.org

[6] Republic of Türkiye (2016). Law No. 6769 on Industrial Property (Sınai Mülkiyet Kanunu). https://www.mevzuat.gov.tr

[7] Mowery, D. C., Nelson, R. R., Sampat, B. N., & Ziedonis, A. A. (2001). The growth of patenting and licensing by U.S. universities: An assessment of the effects of the Bayh-Dole act of 1980. Research Policy, 30(1), 99-119. https://doi.org/10.1016/S0048-7333(99)00100-6

[8] Republic of Türkiye (2001). Law No. 4691 on Technology Development Zones (Teknoloji Geliştirme Bölgeleri Kanunu). https://www.mevzuat.gov.tr

[9] Ries, E. (2011). The Lean Startup. Crown Business.